

Retained-Response Derivative Bounds and a Wider Conditional MQM Thermal Capture Window

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Abstract—We tighten the response-continuity allowance for the frozen MQM thermal capture model. Bath-only evolution disappears under the retained trace, reducing the analytic dyad generator charge from 64 to 14. A new finite-192-mode response derivative calculation, enclosed by parameter and arithmetic allowances, bounds its ideal half-trace norm below 2^{-20} at $T = 16$. A finite-response derivative and curvature bound, combined with uniform continuum value comparisons, certifies retained thermal error below $31/2^{21} < 2^{-16}$ throughout $[16 - 2^{-14}, 16 + 2^{-14}]$ at unchanged $\beta = 5/16$. This quadruples the prior certified halfwidth. A uniform state comparison does not certify a continuum derivative; that numerical derivative transfer remains open. The result is fixed-launch, finite-time and conditional, with no physical promotion or permanent retention claim.

Index Terms—Quantum control, retained response, derivative bounds, thermal capture, timing tolerance.

I. FROZEN CONTRACT

Retain $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The admitted launch is $p0$ and untouched banks/references are identity spectators. The captured target is stationary in the source frame. Model time is not calibrated device time. Mathematical bath modes are not physical qubits.

The parent conditional window used [1]

$$\frac{1}{2}\|\Delta'_2(t)\|_1 \leq 4V_\beta(64t^2 + t), \quad (1)$$

where $V_\beta < \bar{V} = 64/31457265$. It certified halfwidth 2^{-16} . The earlier point-time numerical certificate [2] and Gaussian/continuum dependencies remain assumptions, not external-review conclusions. The target and source parameters are unchanged. This is post-exposure engineering certification, not independent prediction.

II. REMOVING THE BATH-ONLY GENERATOR

Decompose the vacuum reference generator as

$$H_0 = H_S + H_B + H_{SB}, \quad H_{SB} = S^\dagger a(c) + S a^\dagger(c). \quad (2)$$

For the trace-class dyads of the source response,

$$\text{Tr}_B[H_B, X] = 0. \quad (3)$$

Thus the retained final-evolution derivative charges only $H_S + H_{SB}$. The native graph has maximum absolute row sum $1 + \lambda < 7/2$. The source $S = I_C \otimes \sigma_R^-$ is a two-level lowering operator. Splitting the bath into its normalized bright mode

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c/G and orthogonal modes gives Jaynes–Cummings exchange blocks with eigenvalues $\pm G\sqrt{n}$. On total charge $Q \leq 2$, $n \leq 2$, and therefore

$$\|H_{SB}\|_{Q \leq 2} \leq \sqrt{2}G < 21/2, \quad G^2 = 128/\pi < 49. \quad (4)$$

Consequently $\|H_S + H_{SB}\|_{Q \leq 2} < 14$. The eight response dyads and triangular-boundary term now yield

$$\frac{1}{2}\|\Delta'_2(t)\|_1 \leq 4V_\beta(14t^2 + t). \quad (5)$$

This bound is a retained-state statement on the declared response sectors, not an unrestricted thermal Fock-space norm. Exact rational accounting with this analytic improvement alone certifies halfwidth 2^{-15} .

III. FINITE RESPONSE DERIVATIVE CERTIFICATE

The stationary $M = 192$ Gauss–Legendre reference has a finite identity [3]

$$\Delta_{2,M}(t) = \text{Tr}_B[U_2(t)\rho_{1,N}U_2(t)^\dagger] - (\text{tr } N)\rho_{0,M}(t). \quad (6)$$

Here $\rho_{1,N} = |p0\rangle\langle p0| \otimes \Gamma_1(N)$. All 192 one-photon columns are propagated in the 19299-coordinate charge-two sector. The derivative is computed by $Z = -iH_2W$ and $z = -iH_1u$, followed by partial tracing the corresponding dyad derivatives.

Propagation uses the parent's 128 steps of duration $1/8$, degree 48, parameter intervals and binary 64 operation-count model [2]. Let d_H enclose ideal/stored generator mismatch, $e = 16d_H + \epsilon_{\text{prop}}$, ν bound $\text{tr } N$, and d_n bound occupation ℓ_1 error. The derivative debit is

$$d_{\text{jet}} = 2\nu\{64(2e + e^2) + d_H(1 + e)^2\} \quad (7)$$

$$+ 128d_n + 2^{-30}. \quad (8)$$

The final term conservatively covers sparse derivative multiplication and retained readout. It exceeds the source operation-count allowance; no eigensolver accuracy is used in the certificate. Half the entrywise sum of absolute real and imaginary stored derivative entries supplies the matrix upper bound.

Exact fraction arithmetic gives

$$\frac{1}{2}\|\Delta'_{2,M}(T)\|_1 < 5.875 \times 10^{-7} < 2^{-20}. \quad (9)$$

The raw eigenvalue diagnostic is approximately 2.734×10^{-11} ; its smaller value is not the certified ceiling. The finite point response is separately below $3/2^{26}$ after its parameter and arithmetic debits.

Since $\|H\|_{Q \leq 2} < 64$ and the hot/cold launch difference has trace norm at most 2ν , double-commutator differentiation gives

$$\frac{1}{2}\|\Delta''_{2,M}(t)\|_1 \leq 4 \cdot 64^2\nu. \quad (10)$$

With $\nu < 2^{-16}$, integration around T yields

$$\frac{1}{2} \|\Delta_{2,M}(t) - \Delta_{2,M}(T)\|_1 \leq 2^{-20}w + w^2/8 \quad (11)$$

for $|t - T| \leq w$.

IV. UNIFORM CONTINUUM VALUE COMPARISON

A small uniform value discrepancy does not control its derivative. We therefore do not transfer the numerical derivative directly. Instead we extend the centered comparison [4] to $U_* = 16 + 2^{-12}$, covering every predeclared candidate interval. Set $M = 192$ and

$$R = \frac{x^M}{M! [1 - x/(M+1)]}, \quad x = BU_*/4. \quad (12)$$

The response vacuum allowance is $336U_*^3 \bar{V} R < 2^{-30}$. For the distinct stationary finite covariance, the inherited Bose/Gauss proof [2] uses $r = 2^{-21}$, $G^2 < 128/3$, and $y = 8U_* + 25/16$. Its allowance is

$$\eta = \frac{4(128/3)r}{1-r} \frac{y^{384}}{384! [1 - y/385]} \quad (13)$$

$$+ \frac{2(128/3)r^9}{1-r}, \quad 2U_*^2 \eta < 2^{-57}. \quad (14)$$

These are uniform response-value comparisons, with ideal quadrature and exact common covariance handled in separate stages. Combining them with (11) bounds the continuum response throughout the interval without claiming a numerical continuum derivative certificate.

V. FOURFOLD WINDOW

The parent local vacuum-population allowance remains [1]

$$q(t) < E_0 + d_q(w), \quad (15)$$

$$E_0 = 3/2^{18} + 2^{-128}, \quad (16)$$

$$d_q(w) = 49w/1024 + 49w^2. \quad (17)$$

For $U = T + w$, the uniform higher-order Gaussian remainder is $x_U^2/[2(1 - x_U)]$, with

$$x_U = x_{16} + 6\bar{V}(Tw + w^2/2), \quad (18)$$

$$x_{16} = \frac{12433696189517}{58512905602490430}. \quad (19)$$

Theorem 1. *Under the declared fixed-launch reservoir and arithmetic assumptions, retained thermal error is below $31/2^{21} < 2^{-16}$ throughout $[16 - 2^{-14}, 16 + 2^{-14}]$.*

Proof. For $w = 2^{-14}$, exact rational arithmetic verifies

$$E_\beta(t) < E_0 + d_q(w) + 3/2^{26} + 2^{-20}w + w^2/8 \quad (20)$$

$$+ 2^{-30} + 2^{-57} + \frac{x_U^2}{2(1 - x_U)} < 31/2^{21} < 2^{-16}. \quad (21)$$

The uniform continuum comparison holds because $T + w \leq U_*$. The target and temperature are unchanged. $\square$

The halfwidth is four times the previous certificate, relative halfwidth 2^{-18} , and full width 2^{-13} in model time. The 2^{-13} halfwidth does not pass this conservative certificate. That is an upper-bound limitation, not a physical failure or maximum-retention theorem. After the derivative improvement, the cold-population allowance dominates the added terms.

VI. SEPARATE CHECKS AND REMAINING GATES

Independent occupation-basis assembly at $M = 4, 8$ verifies the bath-only commutator trace identity, exchange norm and retained derivative against a polarization implementation. Dense/polarization differences are below 7×10^{-20} ; central finite-difference discrepancies are below 8×10^{-16} . These are same-agent separate implementation controls, not external independent review.

An attempted convenient display ceiling $61/2^{22}$ failed exact comparison; the actual target still passed. The initial audit and failure are retained, with corrected display ceiling $31/2^{21}$. The derivative dyadic ceiling is a reporting bound, not a replacement of the original acceptance rule. No frozen physical specification changed.

External review, numerical continuum-derivative transfer, permanent retention, bath refresh, physical two-local implementation and arbitrary-input diamond accuracy remain OPEN. The capacity allocation proposal stays separate. Physical promotion remains zero.

REFERENCES

- [1] B. Mayes, *A Finite Retention and Timing Window for the Conditional MQM Thermal Capture Certificate*, v0.1, 2026.
- [2] B. Mayes, *A Conditional Numerical Acceptance Certificate for the MQM Thermal Boundary Latch*, v0.1, 2026; with covariance comparison in *MQM Chain Response*.
- [3] B. Mayes, *MQM Second-Order Specification*, v0.1, 2026.
- [4] B. Mayes, *Time-Centered Gaussian Quadrature Embedding for the MQM Vacuum-Transfer Certificate*, v0.1, 2026.